International Labour Organization



Weighting sample data: post-stratification vs calibration

Series of webinars on Strategies to calculate LFS sample weights using Calibration.
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Weighting

- A **sample survey** is an instrument to **make inference** about a **target population** on the bases of information collected on a small subset of it, i.e. the **sample**.

- To produce estimates, **each "observed sample unit" must be given a weight which is "the number of units in the population that it represents".**

- "**Weights should be assigned to make the weighted units represent the population of inference [and all its relevant characteristics] as closely as possible"** *(cfr. Brick and Kalton, 1996, Kalton and Cervantes 2003)*

Weighting

- Any sample designs for household surveys (LFS, HBS, HIES, etc.) may result in **samples with imperfections that can lead to bias** in the results
  - Imperfections include:
    - **Unequal probabilities of selection**
    - **Unit non-response**
    - **Under-coverage of the population**
    - **Unequal coverage of subgroups of population**

Weights can compensate and adjust for some of these imperfections

Estimation: Design weights

- **Design weight** (or base weight) is calculated for each sampled unit to reflect unequal probabilities of selection
  - Calculated as the reciprocal of its probability of selection, in SRS

$$\text{probability of selection} = \frac{n}{N} \qquad \text{design weight} = \frac{N}{n}$$

Theoretical

- used for the **Horvitz-Thompson (HT) estimator**
- In **multi-stage designs**,
  - base weights need to be calculated taking into account the probabilities of selection at each stage
    - e.g. probability of the Enumeration Areas to be selected; of the Households; ...by stratum
- It ensures that **total population estimate matches the total population figures from the frame**, but this is not true for population sub-groups

EXAMPLE 1: Use of design weights in SRS

	Population figures	Respondents to the survey	Weight	Estimate
Total	5,000,000	1,000	5,000	5,000,000

The simplest weight, the same for every unit
5,000,000 / 1,000

The estimate coincides with the population

Producing estimates using auxiliary variables

- However, design weights can be used to produce all the survey estimates, i.e. related to both the target and auxiliary variables (sex, age-group, region, nationality etc.)
- Let's assume that the total of the variable X in the target population is known in advance (SEX: 1=male, 0=female)

$$X = \sum_{k=1}^{N} x_k$$

  - Suppose we use a sample **s** of size **n,** and we get that

Population estimate from the sample

$$\hat{X} = \sum_{k=1}^{n} d_k x_k$$ **is far from**

Population figure from external sources

- Of course, $X = \sum_{k=1}^{N} x_k$
  - **these differences are normal** in sample surveys,
  - but **are not desirable, affect credibility** of an NSO and **produce biased results**

EXAMPLE 2: use of design weights

What happens if we look at the estimates obtained for sub-populations ?

	Population figures	Respondents to the survey	Weight	Estimate
Women	2,500,000	600	5,000	3,000,000
Men	2,500,000	400	5,000	2,000,000
Total	5,000,000	1,000	5,000	5,000,000

Different proportion due to sample /refusals / non-contacts / substitutions etc.

The simplest weight (HT), the same for every unit

These estimates are Unsatisfactory / not desirable, moreover affect credibility of an NSO

From Design Weights to Final Weights

A simple solution exists:
men and women can be weighted separately
(also known as POST-STRATIFICATION)

	Population figures	Respondents to the survey	Weight	Estimate
Women	2,500,000	600	4,167	2,500,000
Men	2,500,000	400	6,250	2,500,000
Total	5,000,000			5,000,000

Different weights for different groups, same for every unit in the group

The estimates are all satisfactory

From Design Weights to Final Weights

Actually, we get the final weight, starting from the design weights and applying the correction factors

	Population figures	Respondents to the survey	Design Weight	Direct Estimate	Correction factor	Final Weight	Estimate
Women	2,500,000	600	5,000	3,000,000	0.833	4,167	2,500,000
Men	2,500,000	400	5,000	2,000,000	1.25	6,250	2,500,000
Total	5,000,000			5,000,000			5,000,000

Correction factors= Population figures / Direct Estimate
(2,500,000 / 2,000,000)

should be as close as possible to 1

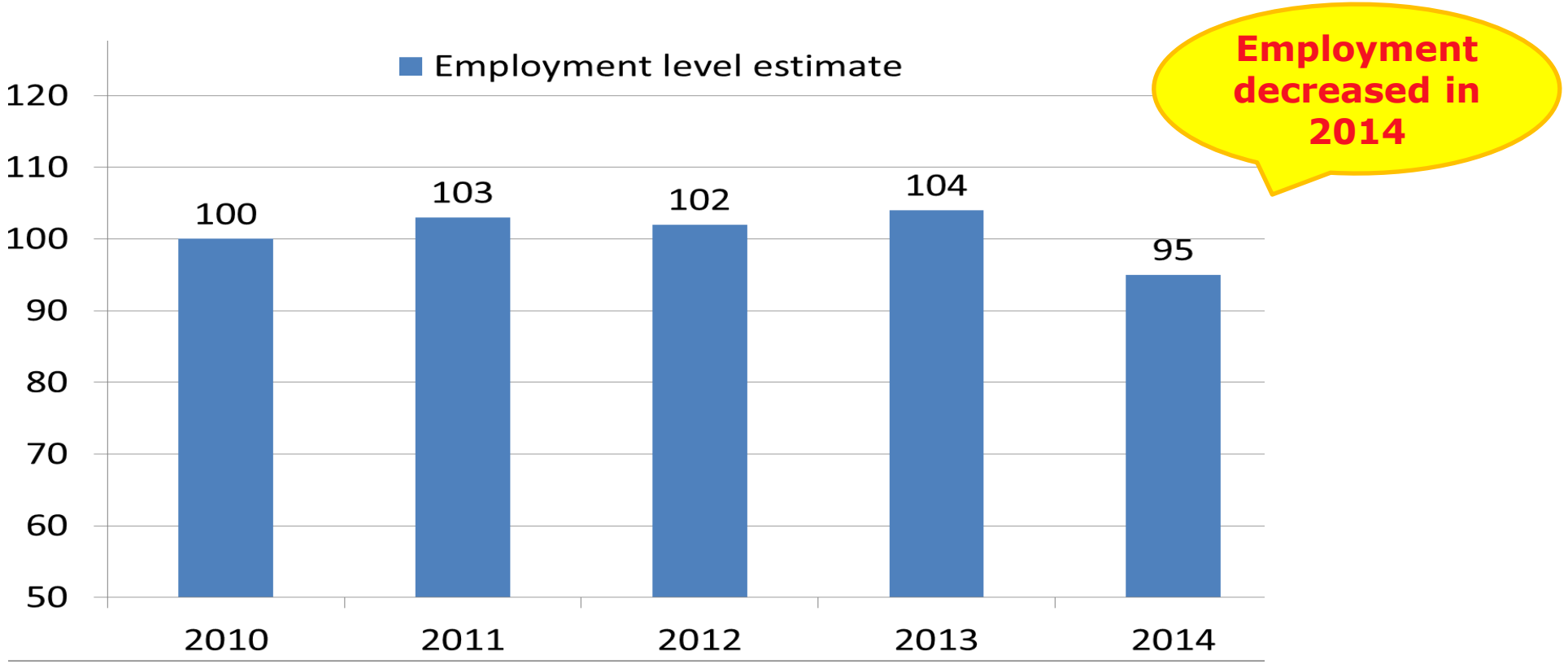
Final weights =
Design Weights * Correction factor

These estimates are all satisfactory
Weighted sample distribution for key variables of interest
(for example, age, sex, nationality, etc.) conforms to a known population distribution.

Eliminating fluctuations of population estimates

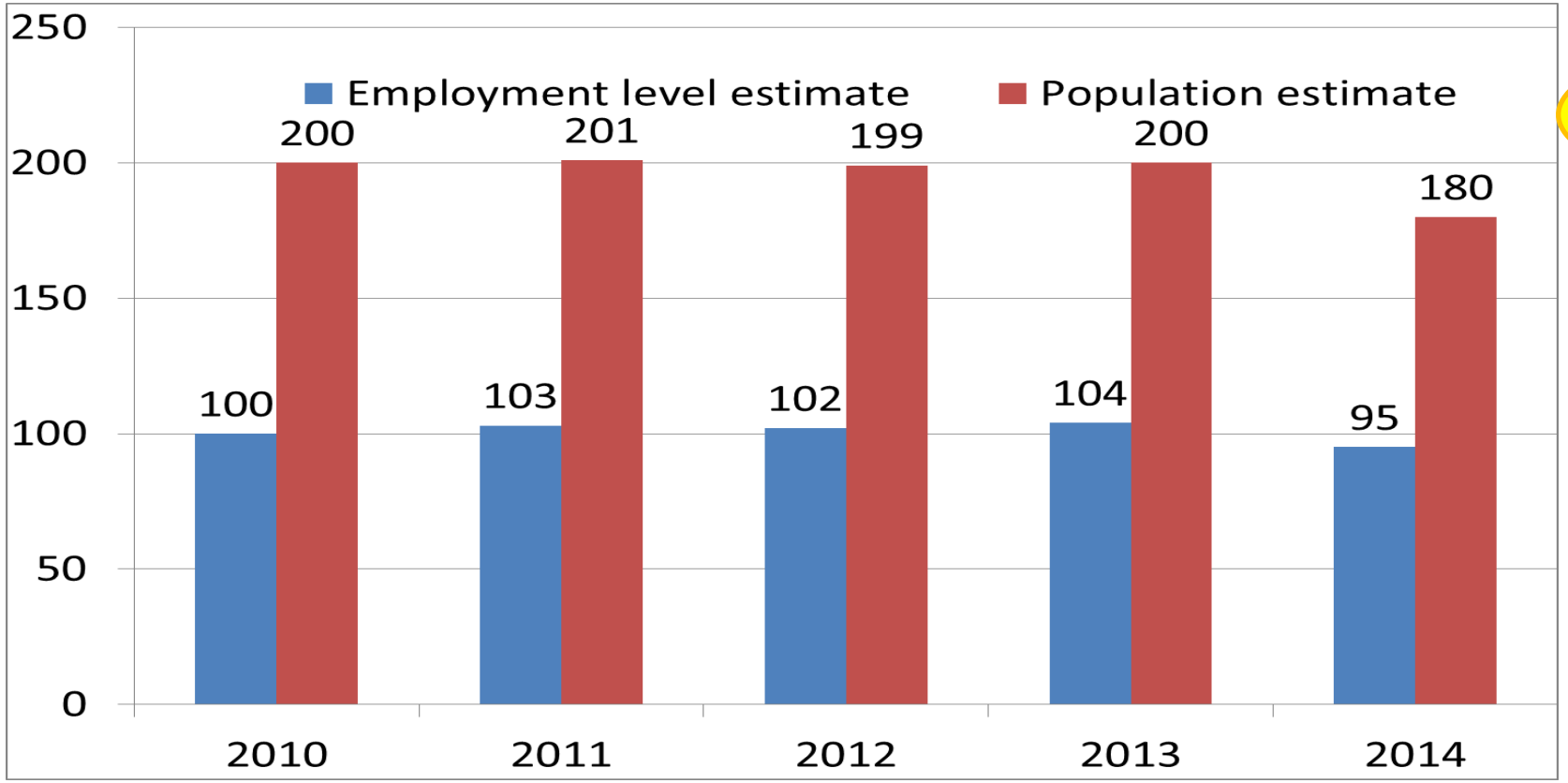
Why ? What do I mean? Let's make an example.

- We are looking at the latest figures on "**Employment of women 15-34 years in rural areas**"
- What can we say about the employment in 2014 ?



Eliminating fluctuations of population estimates

- and what can we say if we look also at the population estimates?
- do we come to the same conclusions about employment?



Do you see something strange in 2014?

Eliminating fluctuations from population estimates

- If we look at the employment to population ratio, are we still convinced of what we saw/said earlier ?

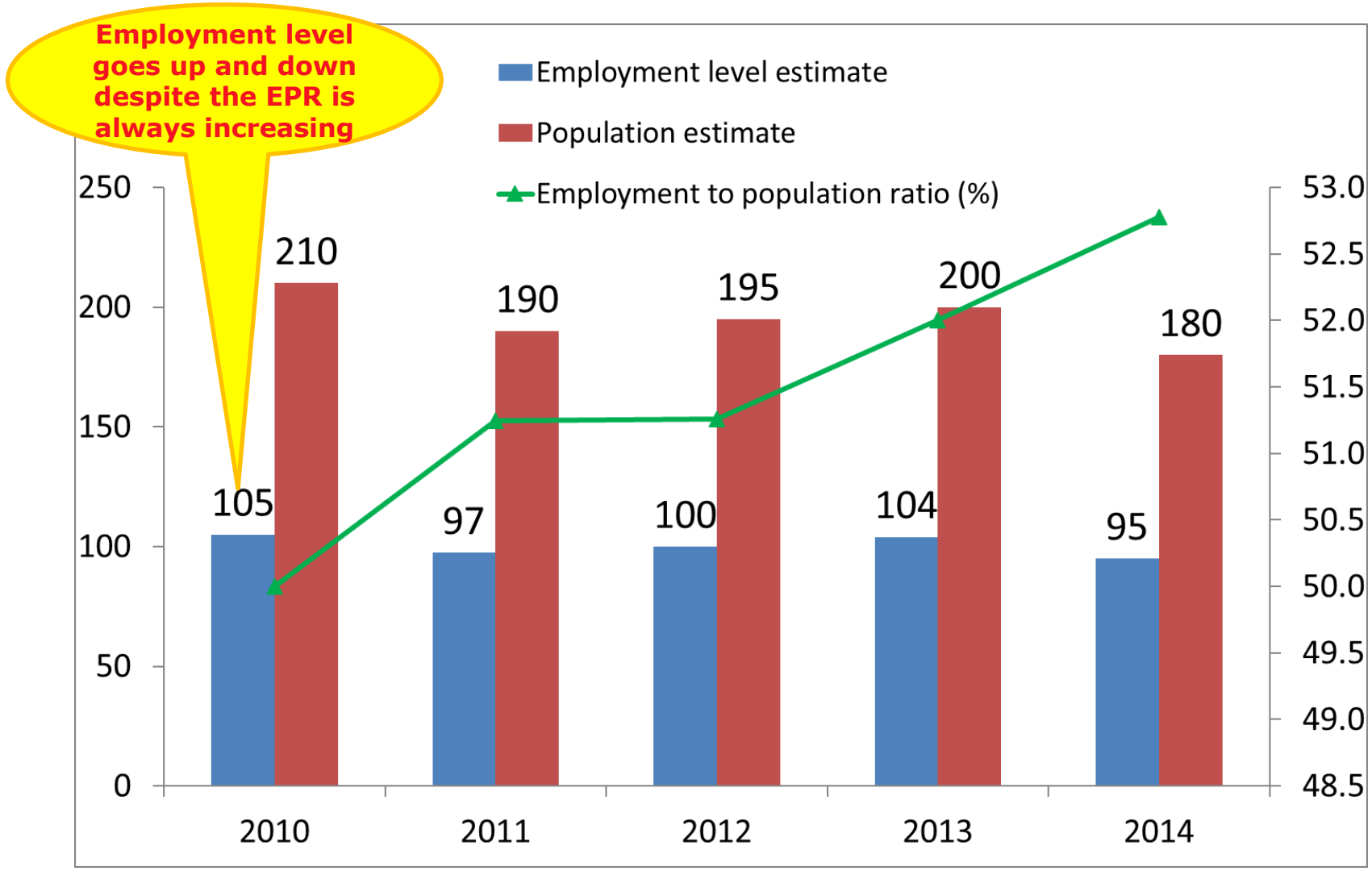
Table1. Employment of women 15-34 years in rural areas					
	2010	2011	2012	2013	2014
Employment level estimate	100	103	102	104	95
Population estimate	200	201	199	200	180
Employment to population ratio (%)	50,0	51,2	51,3	52,0	52,8

Completely different conclusion

- Either there was a “real” big decrease of the women 15-34 years in 2014 or the sample had some “deficiency” and produced biased results, affecting the interpretation of levels
  - The same problem could happen for many other sub-populations: e.g. youth, migrants, etc.
  - Analysts and statisticians have to work together to **decide a minimum set of population figures that should be benchmarked** in calculating weights to avoid such problems

Eliminating fluctuations of population estimates

- In reality, if we do not put benchmarks in the weighting procedure, we risk to have something like the following, where the changes in the employment levels are highly misleading



Reasons for more complex weighting adjustments

- **Design weights** are generally **unsatisfactory** in official statistics
- **Adjustments of design weights** using auxiliary variables correlated with target variable **generally produce more accurate estimates than design weights**
- These adjustments;
  - Help to deal with other important problems, such as:
    - **unit (or total) non-response**
    - **under/over/unequal coverage of subgroups of population**
  - Make **weighted sample estimates conform to population joint or marginal distribution** (consistency with demographic statistics and other surveys)

Reasons for more complex weighting adjustments

- These adjustments;
  - Help to **eliminate fluctuations in population estimates**, thus facilitating the analysis of results
  - Generally produce estimates that are **more precise** and **less biased**
  - **Promote credibility** of NSOs in published statistics.

Post-stratification and **calibration** are the most popular methods for more complex weighting adjustments.

Three-Stage Procedure for Weighting Adjustments in presence of unit non-response

Estimation: Calculation of sample weights

Final weights are usually calculated in a sequence of stages:

- **First Stage**
  - **Design Weight** (inverse of probability of selection)
- **Intermediate Stage**
  - Non-Response Adjustment (total non-response correction factor)
  - **Intermediate Weight** = Design Weight X Non-Response Adjustment
- **Final Stage**
  - Population adjustment (compensation for unequal coverage)
  - **Final Weight** = Intermediate weight X Population adjustment

Some well known weighting methods:

Post-stratification and Calibration

Post-stratification and calibration

Two popular methods to weight sample data taking into account external **auxiliary information to correct the design weights** are:

- **Post-stratification**
  - also known as **cell weighting adjustment**, has been widely used for many years
- **Calibration**
  - is a more complex adjustment method widely used in recent years
  - can incorporate more auxiliary information than post-stratification

It is important that auxiliary variables predict (closely related to) the target variables and response propensity

Post-stratification

Post-stratification:

- **Post-stratification** (stratification after selection of the sample) is carried out
  - dividing the population into **homogeneous groups** (**post-strata**),
    - using joint distribution of auxiliary variables strongly related to characteristics to be investigated (typically sex and age, but also nationality and education level).
  - In each post-stratum, a **unique adjustment factor** is computed and applied to all respondent units to obtain the final weight (**cell-by-cell basis**) .

- It is frequently used also to **reduce bias** due to **non-response** and **under-coverage**

Post-stratification: how it works in practice

	Respondents to the survey			
	0-14	15-74	75+	Total
Women	70	400	130	600
Men	60	250	90	400
Total	130	750	220	1000

Using design weights *d=5000*

	Direct estimates			
	0-14	15-74	75+	Total
Women	350,000	2,000,000	650,000	3,000,000
Men	300,000	1,250,000	450,000	2,000,000
Total	650,000	3,750,000	1,100,000	5,000,000

500,000 / 350,000 = 1.429

	Population figures			
	0-14	15-74	75+	Total
Women	500,000	1,500,000	500,000	2,500,000
Men	500,000	1,500,000	500,000	2,500,000
Total	1,000,000	3,000,000	1,000,000	5,000,000

Calculate correction factors

	Correction factors			
	0-14	15-74	75+	Total
Women	1.429	0.750	0.769	0.833
Men	1.667	1.200	1.111	1.250
Total	1.538	0.800	0.909	1.000

6 adjustment cells

	Final weights			
	0-14	15-74	75+	Total
Women	7143	3750	3846	
Men	8333	6000	5556	
Total				

Using final weights

	Final Estimates			
	0-14	15-74	75+	Total
Women	500,000	1,500,000	500,000	2,500,000
Men	500,000	1,500,000	500,000	2,500,000
Total	1,000,000	3,000,000	1,000,000	5,000,000

6 final weights

Are exactly the same as the population figures from external sources

Post-stratification: characteristics of the estimates

- **Compared to stratification, the variance increases** (the precision decreases) because the actual sample size in each post-strata is not fixed in advance (like in stratification) but is a random variable
  - Would have been better to stratify by sex and age-groups. Anyway, this specific stratification is not usually possible in “households” surveys
- Thus, **post-stratification**
  - **is not efficient like stratification**
  - but is **expected to be more efficient than SRS (producing gains in precision)**
- Using **both stratification of PSUs and post-stratification of FSUs**
  - **is expected to give much better results**

Use of complete multiway distributions

- When we use several variables to post-stratify, strata are derived on the basis of all possible combinations of values **“complete multiway stratification”.**

- If we want to use the following variables
  - **SEX** (2 groups),
  - **AGE** (classified in 14 age-groups),
  - **REGION** (6 groups),
  - **RURAL URBAN AREAS** (2 groups)

we get **2 x 14 x 6 x 2 = 336 adjustment cells**

We need to have sufficient sample units in each of these cells

Use of complete multiway distributions

In this case, post-stratification may have two problems:

- the resulting number of **post-strata** is so high that some of them are **empty in the sample**
  - collapsing strata is a remedy but is time consuming and is not always satisfactory
- **lack of adequate information** about the auxiliary variables or **not detailed enough** to allow complete multiway stratification.
- We may end up using **incomplete multiway stratification**:
  - SEX x AGE GROUP 2 x 14 = 28 adjustment cells
  - REGION x RURAL_URBAN 6 x 2 = 12 adjustment cells
  - SEX x AGE GROUP x REGION 2 x 14 x 6 = 168 adjustment cells

Calibration

Calibration

- has established itself as an **important methodological instrument in large scale production of statistics.**
- is a simple and practical approach to in**corporate a large volume of auxiliary information into the estimation**.
- is used to **produce estimates which are consistent with known aggregates from administrative registers and other accurate sources**
- Implementation is not straightforward. Need to use **software specifically designed to compute calibration weights.**
- Several NSOs have developed **software designed to compute calibration weights**
  - **Some of them also computes standard errors and CVs in a very simple way**

Calibration

- This procedure **adjusts the design weights of each sample unit by "individual" multipliers (i.e. calibration factors)** to obtain final calibration weights that make the estimates agree with known external totals
  - **"individual" multipliers instead of the "cell" multipliers used in post-stratification**
- Particularly **useful when available auxiliary information may consist of several components or layers**
  - regions, districts, provinces, urban/rural localities,
  - persons, households,
  - migrants/non-migrants, etc.
- Particularly **useful in the presence of non sampling errors**
  - frame deficiencies, unequal coverage
  - and as a method for non-response adjustment

Calibration estimators in brief

- The objective of calibration is …
  - to determine new final weights w_k
  - starting from the design weights d_k
  - to satisfy the calibration equation

$$\hat{X} = \sum_{k=1}^{n} w_k x_k \quad = \quad \sum_{k=1}^{N} x_k$$

Population estimates from the sample

Population figures from external sources

then use them to form the calibration estimator of Y

$$\hat{Y}_{CAL} = \sum_{k=1}^{n} w_k y_k$$

All other estimates

Calibration: how it works

- Consider for each element k a positive distance function

$$\boldsymbol{G_k(w_{k,}d_k) \geq 0}$$

- Such that the more the ratio $\frac{w_k}{d_k}$ (i.e. the correction factor) is distant from 1, the higher is the distance (for example $\boldsymbol{(w_k - d_k)^2}$)
- We want to find final weights w_k such that the total distance of all the final weights from the corresponding design weight is minimal

$$\boldsymbol{min}_{w_k} \sum_{k=1}^{n} G_k(w_{k,}d_k)$$

- under the calibration constraints

$$\sum_{k=1}^{n} w_k x_k \quad = \quad \sum_{k=1}^{N} x_k$$

Population estimates from the sample

Population figures from external sources

Post-stratification and Calibration: example 1

Post-stratification

		Population					
		0-14	15-24	25-44	45-74	75+	Total
URBAN	Women	30	80	180	280	80	**650**
	Men	40	90	190	290	90	**700**
RURAL	Women	50	100	200	300	100	**750**
	Men	50	100	200	300	100	**750**
Total		**170**	**370**	**770**	**1170**	**370**	**2850**

Calibration

Adjustment cells with population figures (constraints for weighting using calibration)

URBAN										RURAL									
Men					Women					Men					Women				
0-14	15-24	25-44	45-74	75+	0-14	15-24	25-44	45-74	75+	0-14	15-24	25-44	45-74	75+	0-14	15-24	25-44	45-74	75+
40	90	190	290	90	30	80	180	280	80	50	100	200	300	100	50	100	200	300	100

A total of 20 adjustment cells

The two methods will give the same final weights

Each sample unit is classified in only one cell

Post-stratification and Calibration: example 2

Post-stratification

		Adjustment cells with population figures (constraints for weighting using post-stratification)					
		0-14	15-24	25-44	45-74	75+	Total
NON-MIGRANT POPULATION	Women	350	800	1500	2100	750	5500
	Men	400	900	1700	2500	500	6000
MIGRANT POPULATION	Women	45	95	170	300	30	640
	Men	55	105	230	400	70	860
Total		850	1900	3600	5300	1350	13000

The two methods will give the same final weights

Calibration

Adjustment cells with population figures (constraints for weighting using calibration)

NON-MIGRANT POPULATION										MIGRANT POPULATION									
Women					Men					Women					Men				
0-14	15-24	25-44	45-74	75+	0-14	15-24	25-44	45-74	75+	0-14	15-24	25-44	45-74	75+	0-14	15-24	25-44	45-74	75+
350	800	1500	2100	750	400	900	1700	2500	500	45	95	170	300	30	55	105	230	400	70

Each sample unit is classified in only one cell

Post-stratification and Calibration: example 3

Post-stratification

	COUNTRY LEVEL					
	0-14	15-24	25-44	45-74	75+	Total
Women	395	895	1670	2400	780	6140
Men	455	1005	1930	2900	570	6860
Total	850	1900	3600	5300	1350	13000

COUNTRY LEVEL		Total
NON-MIGRANT POPULATION	Women	5500
	Men	6000
MIGRANT POPULATION	Women	640
	Men	860
Total		13000

Calibration

TOTAL POPULATION										NON-MIGRANT POPULATION		MIGRANT POPULATION	
COUNTRY LEVEL										COUNTRY LEVEL		COUNTRY LEVEL	
Women					Men					Women	Men	Women	Men
0-14	15-24	25-44	45-74	75+	0-14	15-24	25-44	45-74	75+	0+	0+	0+	0+
395	895	1670	2400	780	455	1005	1930	2900	570	5500	6000	640	860

This additional information cannot be used with post-stratification but can be used with calibration

Calibration: much more flexible than post-stratification

Example 4.

COUNTRY TOTAL																URBAN		RURAL	
Men								Women								Men	Women	Men	Women
0-14	15-24	25-34	35-44	45-54	55-64	65-74	75+	0-14	15-24	25-34	35-44	45-54	55-64	65-74	75+	Total	Total	Total	Total

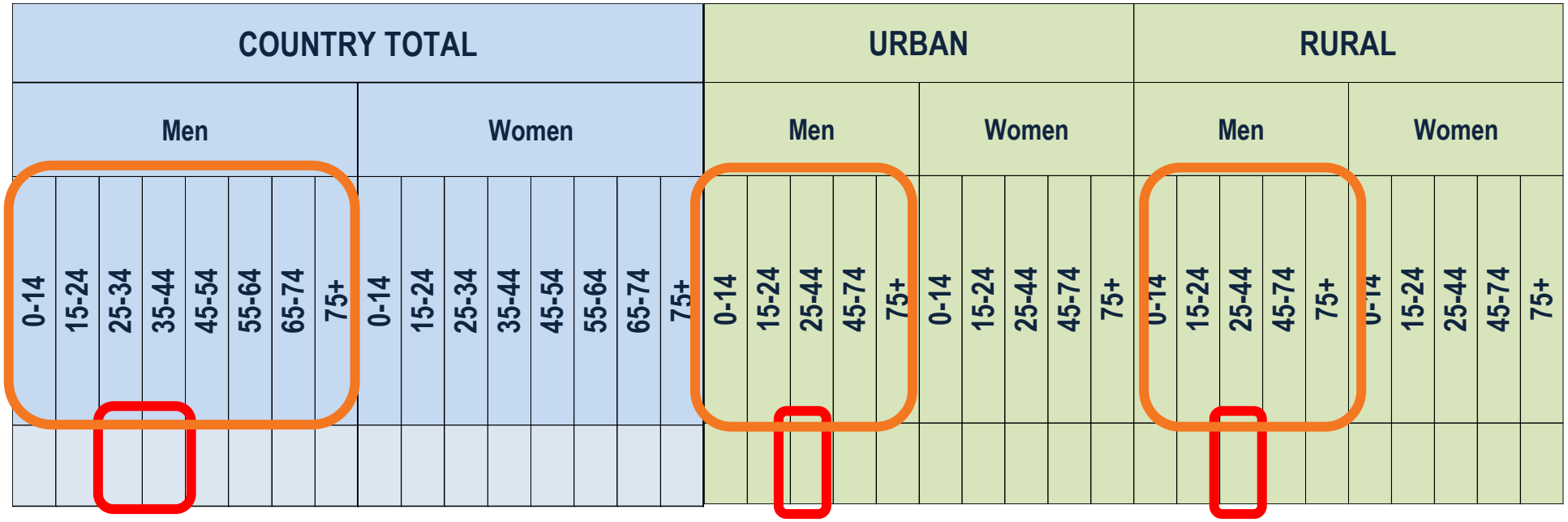
Have to be consistent i.e. the sum of men by age-group = sum of men by urban/rural

age-groups for the total population

No age-groups for urban and rural

Calibration: much more flexible than post-stratification

Example 5.



More age-groups at the country level

fewer age-groups for smaller areas

Calibration: much more flexible than post-stratification

Example 6.

TOTAL POPULATION																MIGRANT POPULATION											
COUNTRY LEVEL																COUNTRY LEVEL											
Women								Men								Women				Men							
0-14	15-24	25-34	35-44	45-54	55-64	65-74	75+	0-14	15-24	25-34	35-44	45-54	55-64	65-74	75+	0-14	15-24	25-64	65+	0-14	15-24	25-64	65+				

more age-groups for the total population

fewer age-groups for the migrant population

Calibration: much more flexible than post-stratification

Example 7.

TOTAL POPULATION								MIGRANT POPULATION							
COUNTRY LEVEL								COUNTRY LEVEL							
Women								Men							
Women								Men							
0-14	15-24	25-34	35-44	45-54	55-64	65-74	75+	0-14	15-24	25-34	35-44	45-54	55-64	65-74	75+
Women				Men				Women				Men			
0-14	15-24	25-34	35-64	65+	0-14	15-24	25-34	35-64	65+	0-14	15-24	25-34	35-64	65+	0-14	15-24	25-34	35-64	65+

more age-groups at the country level

TOTAL POPULATION				MIGRANT POPULATION			
URBAN		RURAL		URBAN		RURAL	
Women	Men	Women	Men	Women	Men	Women	Men
0+	0+	0+	0+				

fewer age-groups for smaller areas or administrative divisions

Quarterly and monthly constraints

Calibration: Quarterly and monthly constraints

Example 9.

| POPULATION OF THE QUARTER | | |

Integrated weights

Calibration: Integrated weights Example 10

- All the members of the same household will have the same final weight
- It is possible to add benchmarks on **household level characteristics** known from external sources
  - No. of households in total, and with native or foreign-born head of household

TOTAL POPULATION		MIGRANT POPULATION		TOTAL POPULATION				MIGRANT POPULATION				HOUSEHOLDS	
COUNTRY LEVEL		COUNTRY LEVEL		URBAN		RURAL		URBAN		RURAL		COUNTRY LEVEL	
Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Women	Men	Non-migrant head of household	Migrant head of household
8 age groups	8 age groups	4 age groups	4 age groups	5 age groups	5 age groups	5 age groups	5 age groups	Total	Total	Total	Total	Total	Total

Population figures from external trusted official sources

Benchmarking to external population figures

- In household sample surveys the external benchmark can be obtained from
  - **Administrative data, population census, civil registers**
  - **Statistical registers** of population build by the NSOs
  - **Population projections**
    - usually based on Population Census, integrated with registration data for births, deaths and migrations.

Population known totals and
the **frame population** must refer
to the same **target population**

How to chose auxiliary information for weighting 2 of 2

- **The benchmark should always be available in all future survey round.**
  - In regular or repeated surveys it is recommended to use the same weighting strategy and the set of constraints (of course with up-to-date figures)
  - If for some reason, the figures are not available in the future, it would not be possible to use the same set of constraints and we could get a break in the related time series.

Lessons learnt 3

- **Design weights are usually unsatisfactory due to sample imperfections**
- **A Three Stage procedure is typically used to get final weights**
  - adjusting for the **unit non-response**
  - accounting for the **undercoverage** and other **sample unbalances**
  - benchmarking to **population figures from external sources**
  - hence **increasing precision and reducing bias** of estimates
- **Post-stratification and Calibration** can be used for all these purposes
- **Calibration much more flexible than post-stratification** when auxiliary information is available for different layers or sub-groups or
  - Entire country, regions, urban/rural,
  - Dwellings, households, persons
  - Migrants/non-migrants, etc.
- **Calibration also useful for sub-modules from a wave approach**

Thank you!